

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Artificial Intelligence Application Development and Jira

PRACTICAL IV

Roll No.: T005	Name: Ritesh Chaubey
Class: TYIT	Batch: 1
Date of Assignment: 25-07-2026	Date/Time of Submission: 25-07-2026

1. Scenario:

A medical test is used to detect a particular disease.

The probability that a randomly selected person has the disease is 1%.

If a person has the disease, the test returns positive with probability 99%.

If a person does not have the disease, the test still returns positive with probability 5% (false positive rate).

A person takes the test and receives a positive result.

Problem Statement:

Write a Python program to compute the probability that the person actually has the disease given that the test result is positive.

Code:-

```
P_D=0.01
P_Pos_D=0.99
P_Pos_NotD=0.05
P_NotD=1-P_D
P_D_Pos=(P_Pos_D*P_D)/((P_Pos_D*P_D)+(P_Pos_NotD*P_NotD))
print("probability that the person actually has the disease is :",round(P_D_Pos*100,2),"%")
```

Output:-

```
probability that the person actually has the disease is : 16.67 %
```